

Assignment 3 - CSE340

1. Subtract -3.0210 from 16.4810 using IEEE-754 single-precision floating-point representation. Check if the result has overflow or underflow or none.

Note: Consider 5 digits while performing the decimal to binary conversion of the floating portion.

2. Answer the following questions.
 - a. How does optimized multiplication improve efficiency and performance compared to traditional long multiplication, especially in terms of speed and computational complexity?
 - b. Why is a bias added to the actual exponent in the IEEE 754 floating-point representation, and how does this affect the encoding of both positive and negative exponents?
3. fle.d x5, f3, f5
 - a. Why is x5 an integer register?
 - b. f3 and f5 both are single precision numbers. True or False. If false, write the correct statement.
 - c. after running the instruction you see that x5 register contains 0. What can you comment on?
4. Write the necessary RISC-V code to compare two floating point registers f1 and f2. If f1 and f2 are equal jump to a label called “**jumpEqual**” else jump to another label called “**jumpNotEqual**”

Note: both the numbers stored in f1 and f2 registers are double-precision floating point numbers.

5. Write the necessary RISC-V code to compare two floating point registers f1 and f2. If f1 is greater than f2, jump to a label called “**jumpGreater**” else jump to another label called “**jumpNotGreater**” and divide the variable c with 16 and store that answer in c.

Note: both the numbers stored in f1 and f2 registers are double-precision floating point numbers and variable c is stored in x19.